

Defect Inspection Report — Routine Defects Notification Period Inspection,
Pier P2 Pier Cap

Nandira River Bridge Project — Package NRB-4

Document Reference:	DIR-NRB4-001
Date:	14-Feb-2024
Contract No.:	NHIA/NRB4/CW/2020-01
From:	QA Engineer, Meridian Engineering Consultants
To:	Resident Engineer (Er. Anand Vasker), copied to project file
Location:	Pier P2, Nandira River Bridge Project — Package NRB-4

Subject: Routine Defects Notification Period Inspection — Pier P2 Pier Cap

This report records the findings of a routine Defects Notification Period inspection carried out 13-Feb-2024, forming part of the programme of periodic inspections during the Defects Notification Period established by Contract Data Section 3 (365 days from the Taking-Over Certificate, 22-Sep-2023 to 21-Sep-2024).

FINDINGS:

- Map (crazing) cracking observed across approximately 60 per cent of the exposed top surface of the Pier P2 pier cap, beneath and around the bearing plinths. Crack widths estimated by visual inspection at between 0.1mm and 0.5mm, in an irregular, interconnected pattern typical of surface shrinkage cracking, not of a single dominant crack or a pattern radiating from a point load.
- No cracking of comparable extent observed on the vertical faces of the Pier P2 pier shaft below the cap, or on the pier cap soffit.
- No spalling, exposed reinforcement, rust staining, or efflorescence observed at any crack. No cracking observed at the bearing plinths themselves or at the bearing seating surfaces.
- This condition was not present, or not of this extent, when Pier P2 was last formally inspected during the Taking-Over inspection (Engineer's Inspection Notes EIN-NRB4-001, 09-Sep-2023), which recorded no structural cracking at any pier.

PRELIMINARY ASSESSMENT:

The pattern, distribution, and shallow appearance of the cracking are consistent with surface (plastic or early-age drying shrinkage) cracking rather than a structural or load-related defect. No immediate safety concern is identified. Given the extent of the cracking, this should be formally notified to the Contractor under the Defects Liability provisions of the Contract for investigation and, if confirmed, remedial action within the Defects Notification Period.

OTHER SITE OBSERVATIONS TODAY:

- Piers P1, P3, P4, P5, P6, P7 and both abutments visually inspected; no comparable cracking or other new defect observed at any other location.
- Outstanding Works Register OWR-NRB4-001 items 1-4 (Scenario 9 punch list) confirmed closed out during previous inspections; not revisited today.

Reported by: QA Engineer, Meridian Engineering Consultants